IBM FileNet P8 V5.5.8

Day 6

Process Engine & Workflow Architecture

Process Engine

Workflow Design

PEDB & Queues

PE Overview | Key Components | Workflow Concepts | Installation

Rajesh Pasham

Learning Objectives – Day 6

PE

Describe Process Engine V5.5.8 architecture and its role in FileNet P8

PE

Identify all PE components: Workflow System, Work Objects, Event Actions, Queues

Concepts

Define work items, step processors, rosters, regions, and workflow routing

Concepts

Explain PE routing types: sequential, conditional, parallel, and iterative

Lab

Install IBM FileNet Process Engine V5.5.8 using IBM Installation Manager

Lab

Create PEDB in DB2 and configure ECMProcessDS JDBC data source in WAS

Config

Create PE_ConnectionPoint in ACCE linking P8Domain to the PE server

Config

Create TrainingRoster and workflow queues in Process Configuration Console

01

Process Engine Overview

Workflow Automation · PE Architecture · Role in FileNet P8 Platform

Process Engine Architecture & Capabilities

PE Architecture Stack

End Users / ICN

Work item claiming, stepping, and reassignment via browser UI

PE REST API

External workflow launch, query, step via HTTP/JSON endpoints

Process Engine Service

Core routing engine. Evaluates expressions, moves work items, enforces deadlines

PEDB (DB2)

Stores all workflow instances, queue items, roster config, audit log

CE Connection

RMI/IIOP port 32768. PE sends/receives document events from CE

Workflow Automation Capabilities

Event-Driven Launch

Auto-launch via OnCheckin, OnCreation CE events

Conditional Routing

Branch on data fields, expressions, user input

Parallel Processing

Split/join for concurrent multi-branch processing

Deadline Management

Auto-escalation when step deadlines expire

Sub-Workflows

Parent workflows launch and wait on child processes

External Integration

Component steps call Java, web services, scripts

02

Key Components

Workflow System · Work Objects · Event Actions · Queues

Process Engine Key Components

Workflow System

PE server (Windows service vwtaskd.exe). Contains all workflow definitions, rosters, queues, and dispatch engine. Port 32768.

Work Objects

Running workflow instances. Each has a unique Work Object Number. Holds F_Subject, custom data fields, milestone, and lifecycle state.

Event Actions

Process Router (auto-launch), Component Step (Java), Script Step (JS), Mailer (email), Abort. Triggered by CE events or workflow steps.

Queues

Two-level: Roster (groups queues, Region, capacity) > Work Queue (FIFO/Custom, participants, deadline expressions, escalation rules).

Work Object Lifecycle

LAUNCHED	IN_QUEUE	IN_PROCESS	STEPPING	COMPLETED
Workflow instance created. F_Subject, data fields set. Work Object Number assigned.	Work item placed in target queue. Waiting for a user to claim it.	User has locked the work item. Actively working on the step.	PE evaluating routing expressions. Transitioning to next step.	All steps executed. Work Object archived to history table.

TERMINATED — Admin or Abort step ends the workflow early.

EXCEPTION — System step error (e.g. component step failure). Requires admin intervention.

Queue Architecture: PE Server > Roster (Region 1) > Work Queue (Participants, Deadline, Max Depth)

`Example: localhost:32768 > TrainingRoster > TrainingApprovalQueue (ceadmin, #AUTHENTICATED-USERS)`

03

Workflow Concepts

Work Items · Step Processors · Workflow Routing · Routing Expressions

Workflow Concepts – Work Items & Step Processors

Work Item vs Work Object

Work OBJECT

Overall workflow instance. Unique Work Object Number.
Persists for the entire workflow lifecycle.

Work ITEM

One step of a Work Object sitting in a queue.
Created when workflow reaches a Queue Step.

Work Item ID = Work Object Number + Step Number

Queue Name — where it's currently waiting

Locked By — who has claimed it (empty = available)

F_Subject — inherited from Work Object

Custom data fields (ApprovalStatus, Priority, etc.)

Step Processor Types

Queue Step	Places work item in queue for human action. Most common.
System Step	Auto-executes. Runs PE scripts, expressions, routing logic.
Component Step	Calls custom Java class (IComponentStep). External systems.
Map Step	Launches sub-workflow. Can wait for child to complete.
Notify Step	Sends email via PE Mailer. No queue assignment.
Abort Step	Terminates workflow immediately. Status: TERMINATED.
Script Step	Executes JavaScript for complex routing logic.

Workflow Routing Types & Expressions

Routing Types

Sequential	Step A → B → C → D in fixed order. No branching. Simplest workflow pattern.
Conditional	Diamond: if ApprovalStatus=Approved go to Publish; else go to Rejection. Branch on data field.
Parallel	Split into multiple branches. Each branch executes concurrently. Join step waits for all to complete.
Iterative	Loop back to earlier step on condition. Example: 3 rejection rounds before auto-terminate.

PE Routing Expression Examples

```
// Simple field comparison
F_ApprovalStatus == "Approved"

// Numeric threshold
F_DocumentCount > 5

// String function
Strings.upper(F_Dept) == "FINANCE"

// Date overdue check
Dates.diff(F_Deadline, Dates.currentDate()) < 0

// Compound logic
F_Priority == "High" && F_Status == "Pending"

// NULL check
F_ApprovedBy != null && F_ApprovedBy != ""
```

04

Installation & Configuration

PEDB Setup · PE Installation · Connection Point · Workflow Queues

PE Installation & Configuration Flow

1. Create PEDB

```
DB2: CREATE DATABASE PEDB PAGESIZE
4096
Grant DBADM to ceadmin
Create PEDATA tablespace
```

2. JDBC Data Source

```
WAS Admin Console:
Create ECMProcessDS
JNDI: ECMProcessDS
Test Connection: Success
```

3. Install PE V5.5.8

```
IBM Installation Manager:
Add PE repository
Set install dir:
C:\IBM\FileNet\ProcessEngine
Configure CE URI: MTOM endpoint
```

4. Bootstrap PEDB

```
Run vwbootstrap.exe
Initializes PEDB schema
Creates PE tables in PEDB
Registers PE with P8Domain
```

5. Start PE Service

```
services.msc:
Start IBM FileNet Process Engine
Verify port 32768 LISTENING
Check vwtaskd.log for errors
```

6. Connection Point

```
ACCE > P8Domain > Process Engine:
New Connection Point:
PE_ConnectionPoint
Server: localhost:32768
Link TrainingOS Object Store
```

Day 6 Lab Exercises

LAB-06-01 · 40 min
Prerequisites & PEDB Setup

- Verify CE and WAS running in ACCE
- Create PEDB in DB2 (PAGESIZE 4096)
- Grant DBADM to ceadmin on PEDB
- Create PEDATA tablespace
- Create ECMProcessDS in WAS — Test Connection

LAB-06-02 · 50 min
Install Process Engine V5.5.8

- Launch IBM Installation Manager
- Add PE V5.5.8 repository
- Install to C:\IBM\FileNet\ProcessEngine
- Run vwbootstrap.exe to initialize PEDB
- Start IBM FileNet Process Engine service

LAB-06-03 · 35 min
Configure PE Connection Point

- Open ACCE > P8Domain > Process Engine
- Create PE_ConnectionPoint (localhost:32768)
- Link TrainingOS Object Store
- Verify Connection Point status: Online
- Test via Process Configuration Console

LAB-06-04 · 35 min
Create Workflow Queues

- Open ACCE Process Configuration Console
- Create TrainingRoster (Region 1)
- Create TrainingApprovalQueue (max: 100)
- Create TrainingReviewQueue (max: 50)
- Add ceadmin as participant to both queues

Day 6 Complete

Process Engine & Workflow Architecture

PE Architecture

Components, lifecycle, capabilities, PEDB, port 32768

Workflow Concepts

Work objects/items, step processors, routing types

Installation

PEDB, ECMProcessDS, IBM IM, bootstrap, queues

Next: Day 7 – Workflow Development using IBM Process Designer